

ILLINOIS VALLEY, IL, USA

WMO: 722149

Lat: 41.352N

Lon: 89.153W

Elev: 654

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04899

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	1.6	-10.2	3.2	-2.4	-5.9	4.1	3.2	28.5	24.0	26.4	26.5	8.7	290	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	91.2	75.1	89.1	74.0	86.4	72.6	78.3	87.5	76.7	85.4	75.1	83.1	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	136.8	83.7	73.7	128.6	81.9	72.4	123.0	80.6	42.3	87.0	40.5	85.7	39.0	83.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.1	23.1	20.0	-10.5	95.7	9.1	4.6	-17.0	99.1	-22.3	101.8	-27.4	104.4	-34.0	107.8	
			DB												
			WB	-11.1	81.6	8.7	1.2	-17.4	82.5	-22.5	83.2	-27.4	83.9	-33.8	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	24.5	26.6	39.6	51.6	63.0	72.0	74.7	72.5	66.9	54.2	41.1	29.7
	DBStd	20.19	12.78	12.03	11.18	9.14	9.01	5.85	5.55	5.26	7.62	9.41	9.84	11.08
	HDD50	2892	791	657	358	90	8	0	0	0	0	60	297	631
	HDD65	5928	1254	1074	790	411	149	9	2	6	67	356	716	1094
	CDD50	3448	2	3	37	136	411	661	767	699	508	192	30	2
	CDD65	1008	0	0	3	8	87	220	304	240	124	22	0	0
	CDH74	9179	0	0	24	121	846	1948	2924	1984	1126	205	1	0
	CDH80	3002	0	0	2	16	268	653	1043	597	370	53	0	0
Wind	WSAvg	9.3	11.0	10.8	11.0	11.6	9.8	8.3	6.6	6.1	7.4	9.3	10.3	10.1
Precipitation	PrecAvg	34.2	1.9	1.6	2.3	3.0	4.0	3.8	3.8	3.9	2.8	2.9	2.6	1.7
	PrecMax	46.2	3.8	5.0	4.8	5.0	7.6	7.8	9.0	10.4	5.8	7.3	5.7	2.8
	PrecMin	20.7	0.3	0.4	0.9	1.3	0.5	1.2	0.1	1.1	0.5	0.4	0.1	0.6
	PrecStd	5.7	0.9	1.1	1.3	1.3	2.1	2.1	2.1	2.3	1.6	1.8	1.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	62.3	77.9	82.1	91.0	93.1	98.0	92.4	91.7	85.9	71.6	60.3
		MCWB	53.0	54.0	64.2	62.3	71.9	75.4	76.9	76.1	74.0	71.1	58.5	57.7
	2%	DB	50.4	54.6	68.9	77.8	86.5	90.0	91.6	88.9	88.3	79.5	64.4	54.0
		MCWB	46.7	49.0	58.8	61.4	68.8	73.3	77.6	75.2	72.4	65.2	56.0	50.6
	5%	DB	45.4	47.1	62.9	73.0	82.4	87.1	88.5	85.9	84.2	74.7	60.8	48.3
		MCWB	42.3	42.7	55.2	58.9	66.0	72.0	75.3	73.5	70.3	63.5	54.1	45.5
	10%	DB	40.8	41.4	56.7	68.2	79.1	83.8	85.8	83.2	81.1	69.8	56.3	44.6
		MCWB	37.9	38.4	49.7	56.4	64.9	70.8	73.5	72.1	67.7	61.2	50.3	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	55.2	65.1	66.4	73.9	80.2	81.0	79.8	77.6	73.0	62.4	58.3
		MCDB	56.8	58.5	76.1	76.8	86.6	89.5	90.7	88.2	89.2	83.2	67.5	59.7
	2%	WB	47.4	49.3	60.2	63.4	71.5	76.0	78.8	77.5	74.6	69.0	58.1	50.7
		MCDB	49.7	55.6	66.6	73.1	81.7	84.9	88.1	86.2	83.0	75.5	62.5	52.5
	5%	WB	42.0	42.9	56.4	60.5	69.3	74.2	77.2	75.7	72.7	65.8	54.5	45.9
		MCDB	45.1	47.0	62.3	70.3	78.8	83.1	86.2	82.7	80.9	71.4	60.3	47.9
	10%	WB	37.9	38.2	50.3	57.9	67.0	72.7	75.5	74.0	70.6	62.3	50.9	41.5
		MCDB	40.8	40.9	56.1	67.1	75.7	81.3	83.9	80.7	78.5	68.5	56.3	44.2
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.1	16.6	20.5	20.3	18.9	18.6	18.2	21.1	19.1	16.1	13.1
		MCDBR	17.8	21.7	23.6	27.8	24.4	22.0	20.2	20.4	23.4	24.6	22.1	18.0
		MCWBR	14.8	16.2	15.3	15.4	11.6	10.3	9.7	9.8	10.3	12.9	15.0	15.5
	5% WB	MCDBR	17.6	20.4	21.7	24.4	21.1	18.2	18.4	17.2	19.3	19.9	19.5	17.2
		MCWBR	15.4	15.9	15.5	15.1	11.3	9.9	10.2	9.4	10.0	13.1	14.7	15.7
Clear-Sky Solar Irradiance	taub	0.288	0.304	0.340	0.392	0.423	0.430	0.448	0.432	0.402	0.347	0.306	0.286	
	taud	2.434	2.396	2.395	2.258	2.225	2.251	2.184	2.225	2.282	2.441	2.504	2.456	
	Ebn at Noon	276	288	288	279	271	269	262	263	263	268	268	267	
	Edh at Noon	25	30	34	42	44	43	46	43	37	29	23	22	
All-Sky Solar Radiation	RadAvg	545	839	1159	1485	1715	1945	1934	1708	1443	959	626	460	
	RadStd	58	102	93	126	126	148	113	102	124	111	54	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page